

In October 2024, Intel and AMD, traditionally fierce competitors in the semiconductor industry, announced the formation of the x86 Ecosystem Advisory Group. This unprecedented collaboration aims to accelerate innovation within the x86 architecture, which has been foundational to computing for over four decades. The alliance is a strategic response to the increasing adoption of ARM-based processors in laptops, desktops, and servers -- a trend that poses a significant challenge to x86's market dominance. The advisory group includes not only Intel and AMD but also major industry players such as Broadcom, Dell, Google, HP E,

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X86 FIGHTS BACK



Lenovo, Microsoft, Oracle, and Red Hat. This broad coalition underscores the industry's commitment to ensuring the x86 architecture remains competitive and relevant. By fostering collaboration among hardware and software devs, the group aims to standardise and enhance x86 implementations, thereby simplifying software development and ensuring interoperability.

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INDIA'S CHIP DREAMS



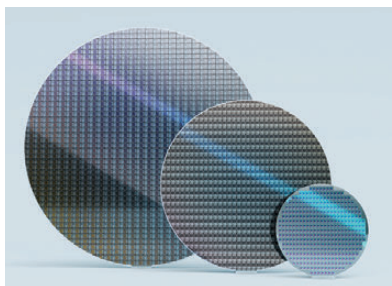
The Indian government has launched the Semiconductor Mission, allocating \$10 billion to attract global semiconductor manufacturers and establish a robust domestic ecosystem. This initiative aims to reduce reliance on imports and position India as a key player in the global semiconductor supply chain. Collectively, these efforts underscore India's chip dreams. The

Tata Group is establishing a semiconductor assembly and test facility in Assam, with operations expected to commence by mid-2025. This facility aims to produce 48 million chips daily, catering to sectors such as automotive, consumer electronics, and telecommunications. Additionally, Micron Technology is developing an assembly and test plant in Gujarat, projected to be operational by early 2025. This facility will feature 500,000 square feet of clean room space dedicated to advanced packaging and testing operations. These developments are expected to create over 300,000 jobs by 2026.

SMC has outlined its roadmap, planning to commence production of its 2nm process, known as N2, in 2025. This node is expected to employ gate-all-around (GAA) transistor architecture, enhancing performance and energy efficiency. Intel is also progressing toward sub-3nm technologies. The company has announced its Intel 20A and 18A nodes, with Intel 20A slated for production in 2024 and Intel 18A in 2025. These nodes will introduce RibbonFET transistors and PowerVia backside power delivery, aiming to improve transistor performance and

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BEYOND 3NM CHIPS



power efficiency. Samsung has been a pioneer in GAA technology, initiating mass production of its 3nm GAA process in mid-2022. They plan to further refine this technology for nodes beyond 3nm, focusing on enhancing transistor performance and reducing power consumption. These advancements will drive innovation across various sectors.